

Flow B - Controlled Wazuh Deployment with Approval Gate and Rollback

A gated deployment workflow for moving validated Wazuh detection content from GitHub into a lab Wazuh manager with evidence and auditability.

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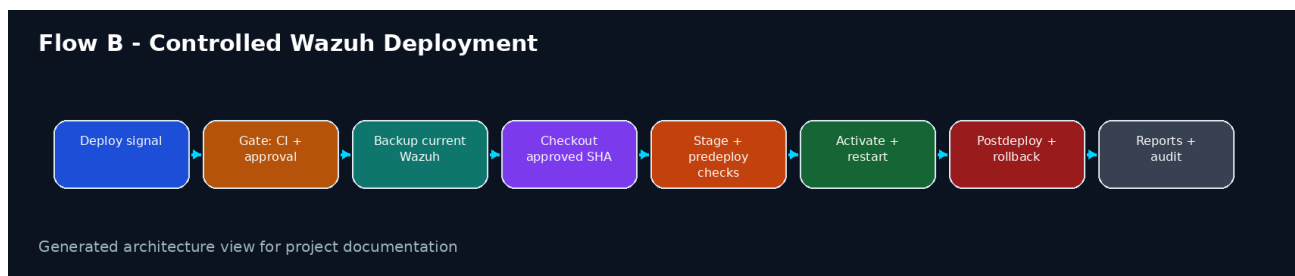
Document Type Workflow project document Flow B	Prototype Focus Controlled deployment Wazuh manager GitHub n8n
Project Character Portfolio MVP with implementation, testing, and validation evidence	Design Goal Demonstrate safe detection deployment with approval gates, validation, rollback, and reporting

Why this document matters

- This write-up works both as a portfolio project document and as an implementation-facing prototype note.
- It explains the problem, the workflow design, the SOC value, and the validation evidence that proves the workflow worked in the lab.
- The screenshots are real build/test artifacts collected from n8n, GitHub, Wazuh, Slack, TheHive, and n8n DataTables.

Key outcomes

- Requires CI pass, ready-to-deploy, approval, and a valid deployment signal before changes are applied.
- Performs backup, checkout, staging, predeploy XML and smoke checks, activation, restart, postdeploy test, and rollback path.
- Reports deployment status to GitHub and Slack and writes deployment run evidence into n8n DataTables and dashboard metrics.



Architecture view - Flow B converts an approved GitHub signal into a controlled Wazuh deployment lifecycle.

1. Project Summary

Flow B is the release-control layer of the prototype. It prevents detection content from being pushed into the Wazuh manager until the pull request satisfies the required gate conditions. It then performs a controlled deployment sequence with validation, restart, postdeploy testing, and rollback capability.

2. Problem Statement

- Validated detection content still needs a controlled and auditable path into the Wazuh manager.
- A manual deployment can skip backup, staging, smoke testing, or rollback preparation under time pressure.
- SOC teams need a deployment history that explains who triggered the change, which SHA was used, and what happened.

3. Gate Conditions and Deployment Lifecycle

Valid signals	ready-to-deploy label, approved review, /deploy-lab comment, or merge-to-main signal
Required gate	detection-ci-pass + ready-to-deploy + approved + valid deploy signal
Safety steps	Backup current Wazuh content, checkout approved commit, stage rules/decoders, XML check, smoke logtest
Activation path	Activate content, restart Wazuh manager, run postdeploy test
Failure handling	Rollback from backup when postdeploy health fails
Audit outputs	GitHub deployment report, Slack status, flow_b_deployment_runs, SOC dashboard metrics

4. Implementation Evidence



Figure B1 - Full Flow B n8n workflow canvas showing deploy trigger, gate, deploy steps, rollback, reporting, and audit table output.

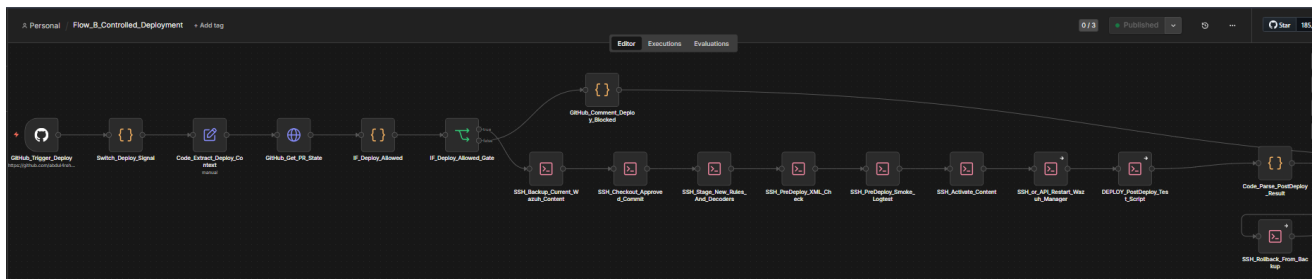


Figure B2 - Flow B left-side zoom: GitHub trigger, deployment signal parsing, PR state lookup, and gate decision.

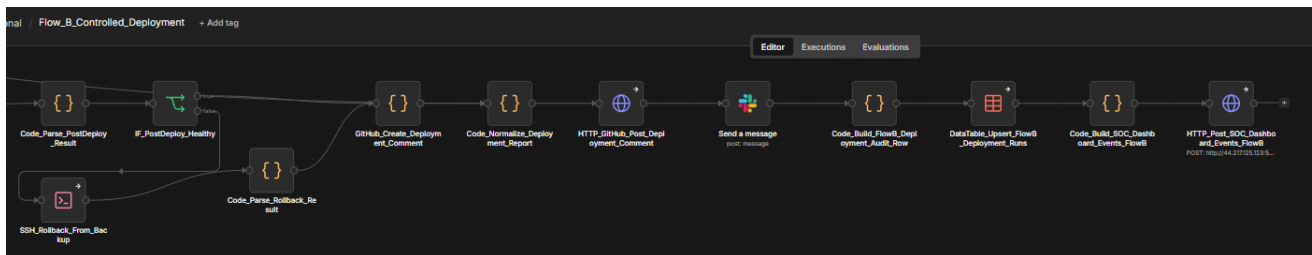


Figure B3 - Flow B right-side zoom: reports, Slack notification, deployment audit table, and dashboard posting.

```

GNU nano 7.2 .env.ci *
# ----- Flow B deploy settings -----
FLOWB_TMP_DIR=/tmp/flowb
FLOWB_REPO_DIR=/tmp/flowb/repo
FLOWB_MANIFEST_PATH=/tmp/flowb/deploy_manifest.json
FLOWB_BACKUP_FILE=/tmp/flowb/last_backup_path.txt

DEPLOY_WAZUH_MODE=remote_ssh
DEPLOY_WAZUH_HOST=#####
DEPLOY_WAZUH_USER=root
DEPLOY_WAZUH_PORT=22
DEPLOY_SSH_KEY="/opt/detection-ci/wazuh-genai-ci/.runtime-ssh/n8n_to_wazuh_ed25519"

DEPLOY_WAZUH_RULES_DIR=/var/ossec/etc/rules
DEPLOY_WAZUH_DECODERS_DIR=/var/ossec/etc/decoders
DEPLOY_WAZUH_STAGE_DIR=/var/ossec/tmp/deploy-flowb/stage
DEPLOY_WAZUH_BACKUP_DIR="/root/wazuh-flowb-backups"

DEPLOY_WAZUH_RESTART_CMD="systemctl restart wazuh-manager"

DEPLOY_API_URL=https://#####:55000
DEPLOY_API_USERNAME="wazuh"
DEPLOY_API_PASSWORD="#####"
DEPLOY_API_VERIFY_TLS=false

```

Figure B4 - Deployment environment configuration context used by shell scripts.

```

root@n8n:/opt/detection-ci/wazuh-genai-ci# ls -lh scripts/deploy/
total 40K
-rwxr-xr-x 1 ubuntu ubuntu 2.3K Apr 23 20:00 activate_content.sh
-rwxr-xr-x 1 ubuntu ubuntu 963 Apr 23 18:16 backup_current_wazuh_content.sh
-rwxr-xr-x 1 ubuntu ubuntu 3.1K Apr 23 18:26 checkout_approved_commit.sh
-rwxr-xr-x 1 ubuntu ubuntu 1.9K Apr 23 18:26 common.sh
-rwxr-xr-x 1 ubuntu ubuntu 4.0K Apr 23 18:05 postdeploy_test.py
-rwxr-xr-x 1 ubuntu ubuntu 2.1K Apr 23 18:31 predeploy_smoke_logtest.sh
-rwxr-xr-x 1 ubuntu ubuntu 1.6K Apr 23 18:31 predeploy_xml_check.sh
-rwxr-xr-x 1 ubuntu ubuntu 583 Apr 23 18:04 restart_wazuh_manager.sh
-rwxr-xr-x 1 ubuntu ubuntu 1.1K Apr 23 18:34 rollback_from_backup.sh
-rwxr-xr-x 1 ubuntu ubuntu 2.0K Apr 23 18:26 stage_new_rules_and_decoders.sh

```

Figure B5 - Deployment scripts present in the repository for backup, checkout, staging, validation, restart, postdeploy, and rollback.

The screenshot shows a Node.js application interface with three main sections: INPUT, COMMAND, and OUTPUT.

INPUT: A JSON schema for the input data, including fields like "exitCode", "stderr", "stdout", "stage", "backup_current_content", "status", "pass", "backup_path", "root/wazuh-flowb-backups/flowb_2026042411304252.tgz".

COMMAND: A shell script snippet that runs a checkout operation on the approved commit, using the command: `cd /opt/detection-ci/wazuh-genai-ci && bash scripts/deploy/checkout_approved_commit.sh`.

OUTPUT: A JSON response showing the result of the checkout operation, including fields like "exitCode", "stderr", "stdout", "stage", "checkout_approved_commit", "status", "pass", "base_sha", "manifest_path", "deployable_count", "wazuh_rules", and "wazuh_decoders".

Figure B6 - Checkout approved commit node output showing deployable Wazuh rule content.

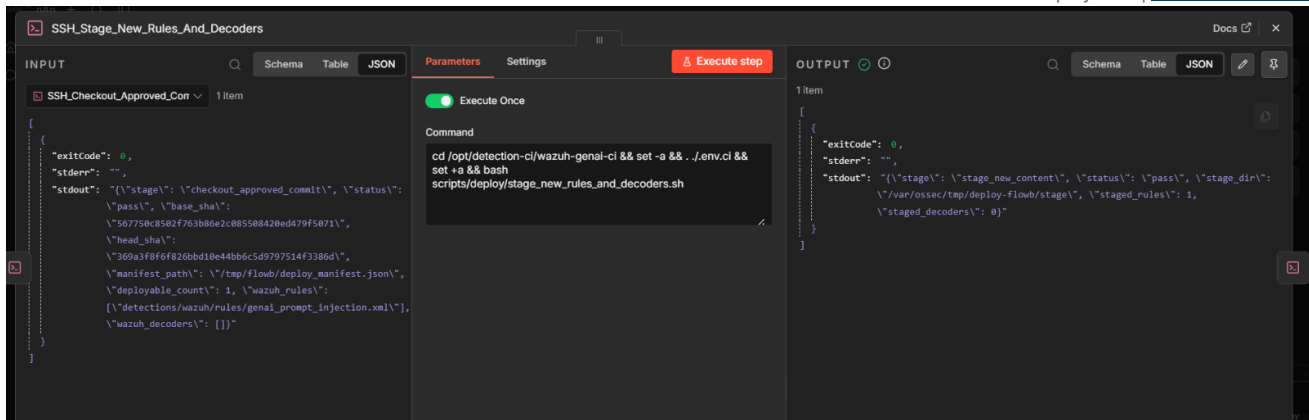


Figure B7 - Stage new rules and decoders node output showing Wazuh staging path and staged counts.

5. Validation and Testing Evidence

Test B1 - Blocked deployment gate

- A deployment command was sent against a PR that did not satisfy the gate.
- The workflow refused deployment and explained the missing requirements instead of touching the Wazuh manager.



Figure B8 - GitHub PR #26 showing /deploy-lab and Flow B BLOCKED report.

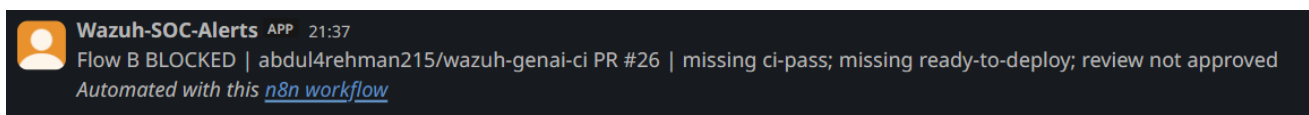


Figure B9 - Slack notification showing Flow B BLOCKED and the missing gate requirements.

The screenshot shows the configuration and output of the `IF_Deploy_Allowed_Gate` step. The **INPUT** section displays a JSON object with deployment details. The **Parameters** section shows the condition `{{ $json.deploy_allowed }}` set to `is true`. The **OUTPUT** section shows the result for the **False Branch (1 item)**, where `deploy_allowed` is `false` and a `deny_reason` is provided.

```

INPUT
IF_Deploy_Allowed 1 item
{
  "owner": "abdul4rehman215",
  "repo": "wazuh-genai-ci",
  "pr_number": "26",
  "issue_number": "26",
  "head_sha": "712e53d2933ba296e07688af464fba27d7bad67",
  "base_sha": "567758c8502f763b88e2c085588420ed479f5071",
  "base_ref": "main",
  "actor": "abdul4rehman215",
  "deploy_signal": "manual_deploy_comment",
  "event_name": "issue_comment",
  "comment_body": "/deploy-lab",
  "review_state": "",
  "pr_url": "https://github.com/abdul4rehman215/wazuh-genai-ci/pull/26",
  "labels": [
  ],
  "merged": false,
  "deploy_allowed": false,
  "deny_reason": "missing ci-pass; missing ready-to-deploy; review not approved"
}

Parameters
Conditions
{{ $json.deploy_allowed }} to is true
Convert types where required
Options

OUTPUT
True Branch False Branch (1 item)
{
  "owner": "abdul4rehman215",
  "repo": "wazuh-genai-ci",
  "pr_number": "26",
  "issue_number": "26",
  "head_sha": "712e53d2933ba296e07688af464fba27d7bad67",
  "base_sha": "567758c8502f763b88e2c085588420ed479f5071",
  "base_ref": "main",
  "actor": "abdul4rehman215",
  "deploy_signal": "manual_deploy_comment",
  "event_name": "issue_comment",
  "comment_body": "/deploy-lab",
  "review_state": "",
  "pr_url": "https://github.com/abdul4rehman215/wazuh-genai-ci/pull/26",
  "labels": [
  ],
  "merged": false,
  "deploy_allowed": false,
  "deny_reason": "missing ci-pass; missing ready-to-deploy; review not approved"
}

```

Figure B10 - IF_Deploy_Allowed_Gate output proving the false branch with deny reason.

Test B2 - Approved deployment success path

- A gated PR with approved, detection-ci-pass, and ready-to-deploy labels was used for the allowed path.
- The workflow completed backup, checkout, staging, validation, activation, restart, postdeploy testing, GitHub reporting, Slack notification, and audit table update.

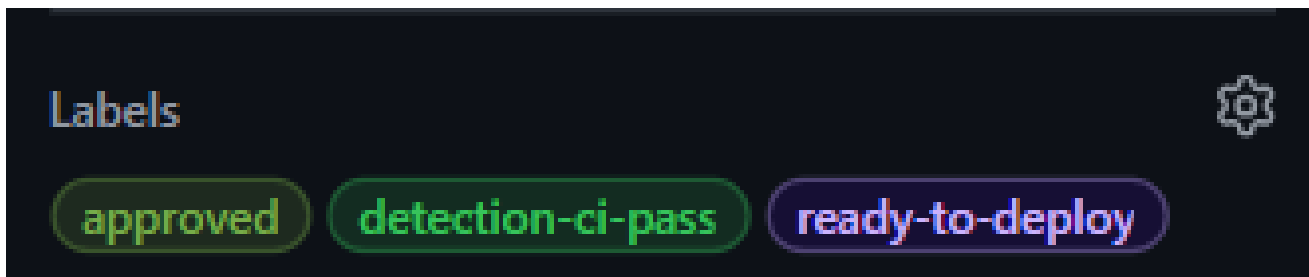


Figure B11 - Required GitHub labels present: approved, detection-ci-pass, ready-to-deploy.

abdul4rehman215 commented now Owner Author ...

/deploy-lab

Flow B Deployment Report

Result: SUCCESS
PR: [#25](#)
Actor: abdul4rehman215
Head SHA: [369a3f8](#)
Repo: abdul4rehman215/wazuh-genai-ci

Deployable Content

- Wazuh rules: 1
- Wazuh decoders: 0
- Deployable count: 1

Stage Results

- Stage to remote manager: pass
- Predeploy XML check: pass
- Predeploy smoke logtest: pass
- Activate content: pass
- Restart manager: pass
- Postdeploy test: pass

Figure B12 - GitHub deployment report showing **SUCCESS** and stage results.

Wazuh-SOC-Alerts APP 18:34

Flow B SUCCESS | abdul4rehman215/wazuh-genai-ci PR #25 | failed: none

Figure B13 - Slack notification showing Flow B **SUCCESS**.

Data tables / flow_b_deployment_runs |

T deployment_id	T run_id	T workflow_execution_id	started_at	completed_at	T environment	T repo_name	T repo_owner	# pr_number	# issue_number	T actor	T event_name	T deploy_signal
flowbwazuh-genai-ci-pr-267265322	flowbwazuh-genai-ci-pr-267265322	149	2026-04-25T21:38:21.0	2026-04-25T21:38:21.0	lab	wazuh-genai-ci	abdul4rehman215	26	26	abdul4rehman215	issue_comment	ignore
flowbwazuh-genai-ci-pr-27d8af71c27	flowbwazuh-genai-ci-pr-27d8af71c27	154	2026-04-25T21:52:44.1	2026-04-25T21:52:44.3	lab	wazuh-genai-ci	abdul4rehman215	27	27	abdul4rehman215	pull_request	label_ready_to_deploy

Figure B14 - flow_b_deployment_runs table showing blocked and allowed deployment runs.

personal / Data tables / flow_b_deployment_runs |

T deploy_signal	T head_sha	T base_sha	T base_ref	merged	T labels_cnv	deploy_allowed	T deny_reason	T backup_status	T checkout_status
ignore	72e53d933ba296e0788a4644bcca	567780d5027f3b86e2c085508420e	main	☐	Empty	☐	no valid deploy signal, missing ci-pass	Empty	Empty
label_ready_to_deploy	d8af71c27228590a376e41e5d03d8f0	567780d5027f3b86e2c085508420e	main	☐	detection-ci-pass_ready-to-deploy.app	☒	Empty	pass	pass

Figure B15 - flow_b_deployment_runs gate and status columns showing labels, deploy_allowed, backup, and checkout status.

personal / Data tables / flow_b_deployment_runs |

T rollback_status	T rollback_reason	T github_comment_status	T github_comment_url	slack_notified	T slack_channel	# slack_ts	T wazuh_manager_host	T comment_checkout_path	T report_markdown
n/a	Empty	pass	https://github.com/abdul4rehman215	☒	COABJLJAT0	1777 133 300.960219	wazuh	/api/detection-ci/wazuh-genai-ci	## Flow B Deployment Report
n/a	Empty	pass	https://github.com/abdul4rehman215	☒	COABJLJAT0	1777 134 164.262089	wazuh	/api/detection-ci/wazuh-genai-ci	## Flow B Deployment Report

Figure B16 - flow_b_deployment_runs reporting columns showing GitHub comment, Slack status, manager host, and report Markdown.

6. SOC Value and Impact

- Reduces deployment risk by requiring explicit approval and validated CI results before content activation.
- Improves MTTR for detection rollout because staging, smoke testing, and restart actions are automated and repeatable.
- Provides traceability for who deployed what, from which PR/SHA, and with which postdeploy result.
- Creates a reusable release model for Wazuh detection content beyond this GenAI use case.

7. Skills Demonstrated

- Deployment automation
- GitHub PR gate design
- Wazuh content staging and validation
- Rollback-aware workflow design
- n8n DataTable audit modeling
- Slack/GitHub operational reporting

8. Prototype Limitations and Future Enhancements

- The lab uses a single Wazuh manager and controlled shell scripts rather than a multi-environment release platform.
- Future versions could add canary managers, approval via GitHub environments, signed deployment manifests, and automated rollback evidence screenshots.
- Deployment table metrics can later power release dashboards and change-control review.

9. Conclusion

Flow B demonstrates the operational bridge between validated detection content and a live Wazuh manager. It proves that the MVP can deploy safely, block unsafe requests, and preserve evidence.